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ETE 3351
Lab 5 part 3

Rigol Spectrum_Analyzer_part2_waveforms

Introduction

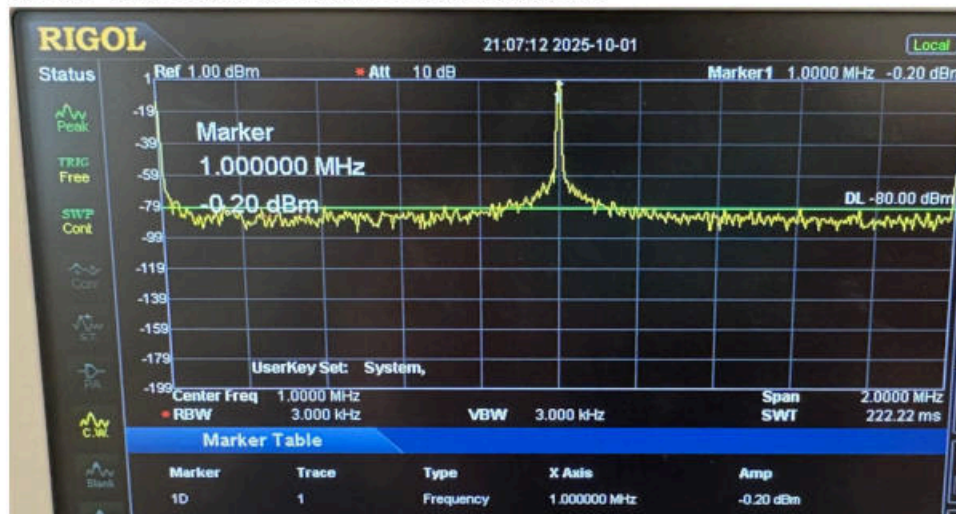
In this lab we explored how electronic signals behave in both the time domain and the frequency domain and how tools like oscilloscopes and spectrum analyzers reveal different aspects of waves and their form. Using the Spectrum Analyzer, we observed how a signal's energy spreads across frequencies allowing us to clearly see the fundamental and its harmonics. We also reinforced key concepts during a laboratory discussion regarding decibels (dB) which express ratios of power on a logarithmic scale and dBm, which is an absolute power level referenced to 1 mW. Logarithmic units are especially useful because they allow us to add changes in gain or loss instead of multiplying them, making signal analysis much simpler.

To better understand how signals sound in real life, we can connect a 45-ohm speaker (Regular 8-ohm speakers require too much current for the signal generator and therefore a higher impedance speaker is preferable). This allowed us to hear the difference between waveforms with different harmonic content, such as the clean tone of a sine wave.

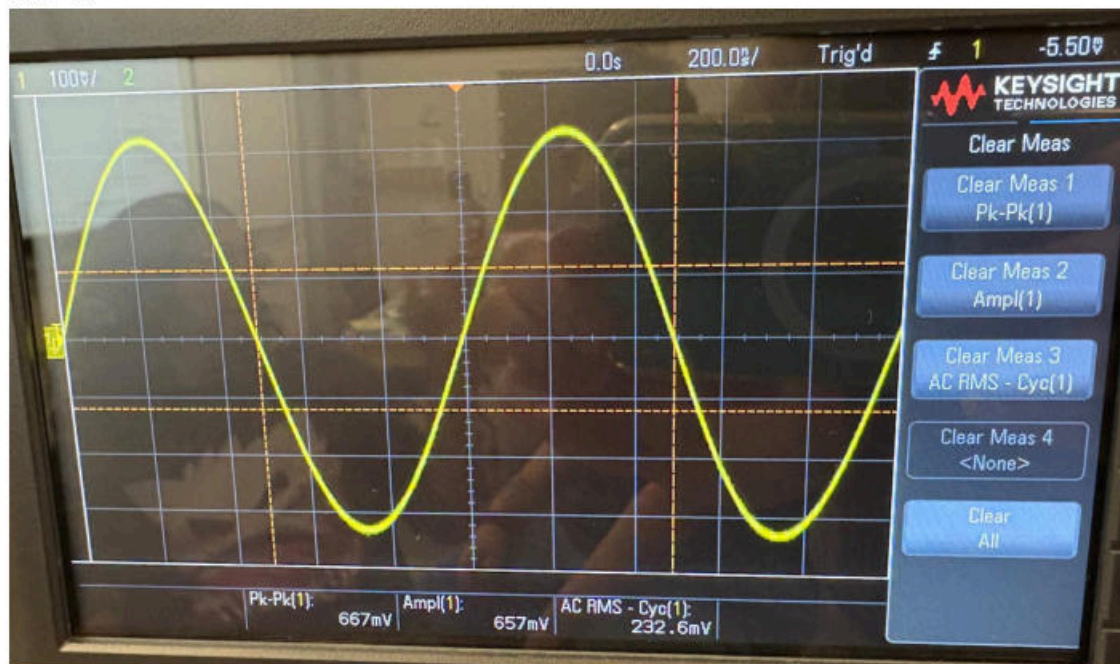
The lab also introduced concepts related to sampling, aliasing, and the Nyquist Theorem. When a signal is sampled too slowly (below twice its frequency) the measured frequency appears incorrectly on the display thus producing an alias. This is why oscilloscopes and analyzers must sample at least $2\times$ the signal frequency to capture it accurately, and ideally $10\times$ or more to reproduce the correct waveform shape (but the more the better). These principles are critical to consider for this lab as well as future labs.

Overall this lab brought together the concepts of waveform shape, harmonics, sampling, decibels, and power measurement. By analyzing both sine and pulse waves we were able to connect theoretical Fourier Series predictions to real measurements, preparing us for more advanced RF and communication system analysis

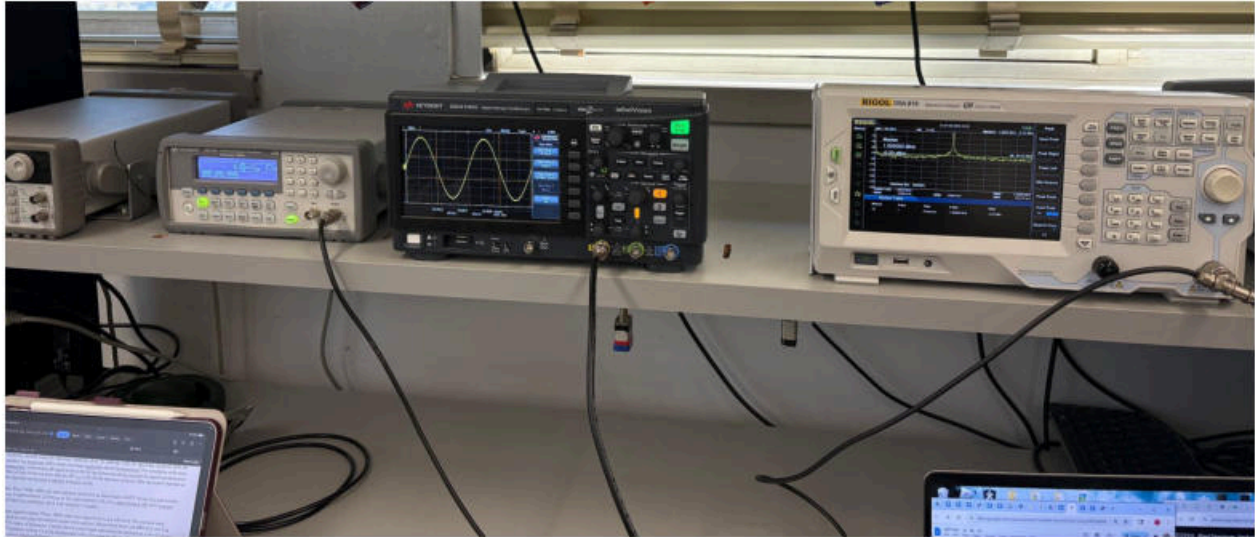
1. Sine Wave 1MHz, 0dBm see intro handout.



IMG 1.



IMG 2.



IMG 3.

Switch the Spectrum Analyzer to linear mode (AMPT → Scale → Lin) and confirm that $V_{rms} \approx 223 \text{ mV}_{rms}$ or $V_{pk} \approx 316 \text{ mV}_{pk}$.

Do not proceed until these values are confirmed — check with the instructor if needed.

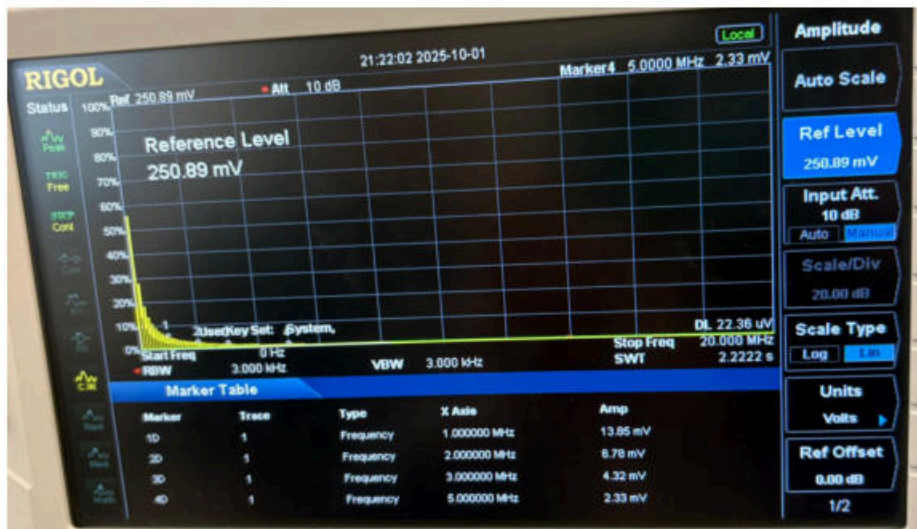
We are trying to reach 0 dBm, which corresponds to approximately 316 mVpp.

Because the function generator and spectrum analyzer are both 50 Ω , the voltage drops due to the 50- Ω loading. The oscilloscope must be disconnected since its High-Z load will affect the reading.

Goal: Adjust the function generator until the Spectrum Analyzer reads 0 dBm ($\approx -6.08 \text{ dBV}$).

To compensate for losing half the signal, Vpp must be doubled to approximately 632 mVpp.

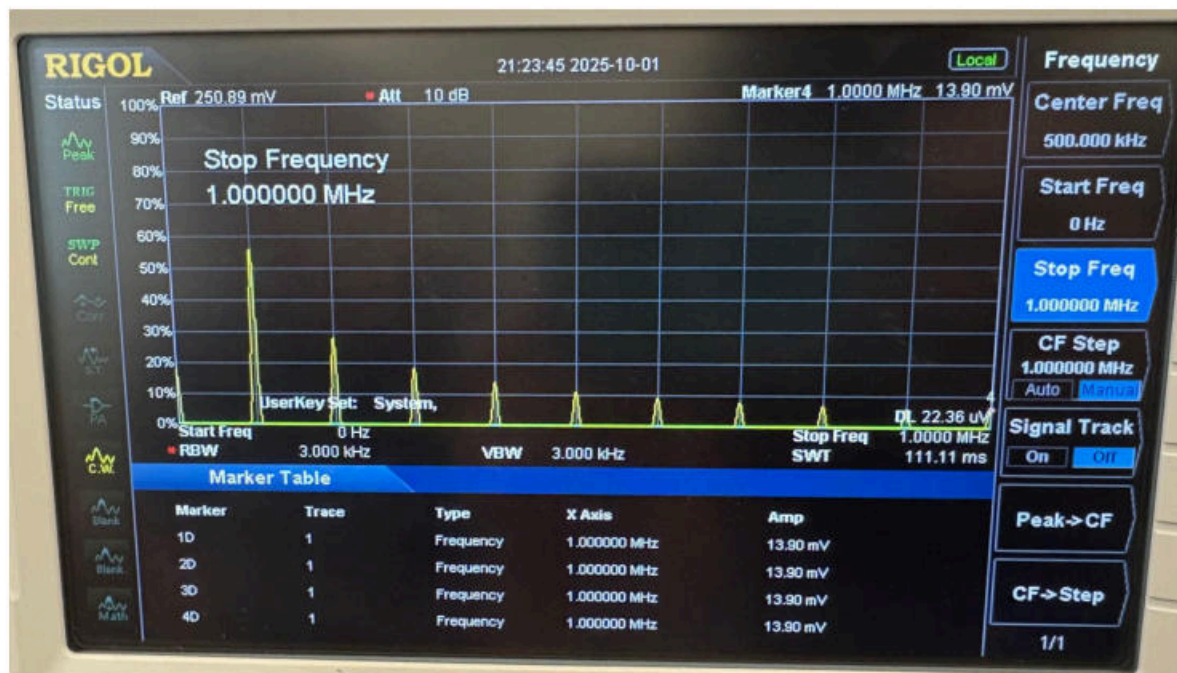
Then measure V_{rms} once the level is correct.



IMG 4. Linear



IMG 5 Log



IMG 6. Zoomed in to 1 mhz range.

2. Change signal to Square Wave, 1MHz make sure signal level is at a safe level, NO overload tones allowed see next page for standard square wave analysis. Record both linear and dBm (Lin and Log AMPT) values of harmonics.

We had to change frequency from 2mhz to 20mhz for max display so we can see the rest of the harmonics.

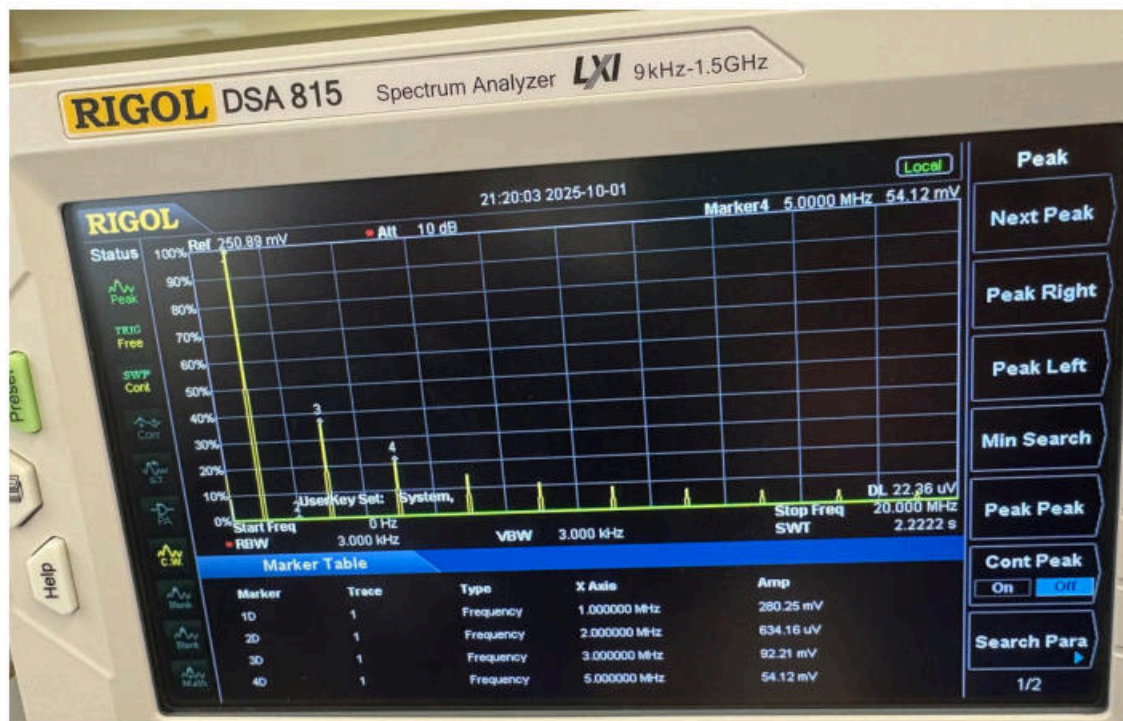
Markers:5

Check $287/5 = 55.6$

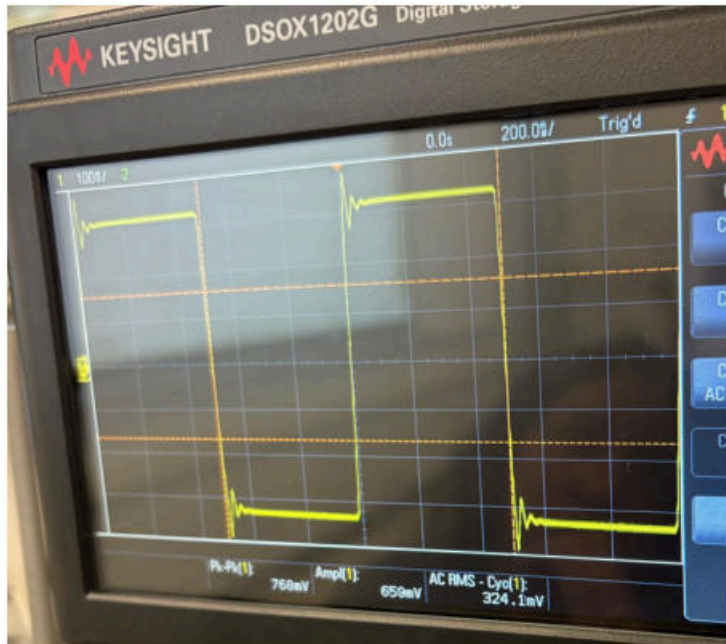
5th s 54.22

Close

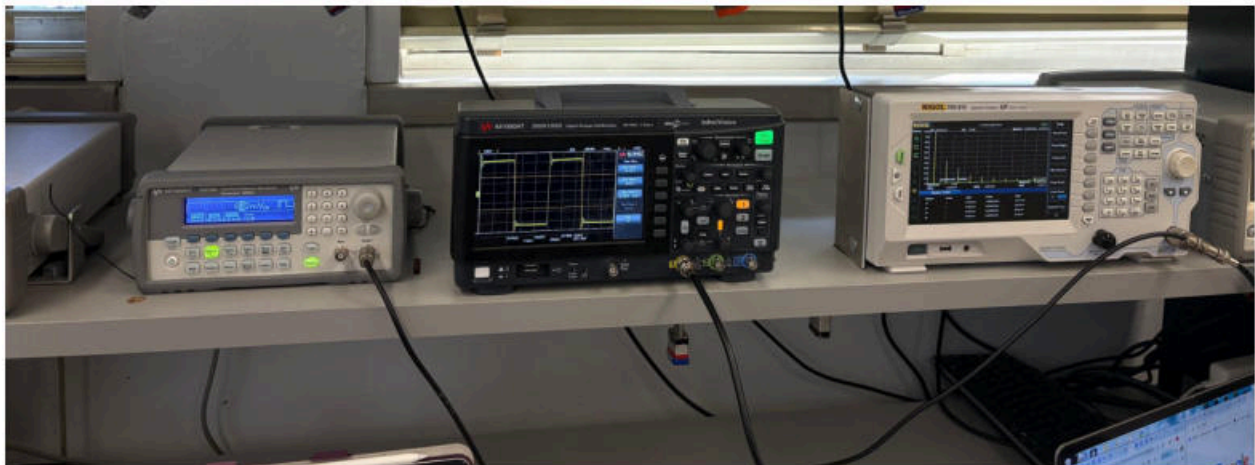
Linear



IMG 7.

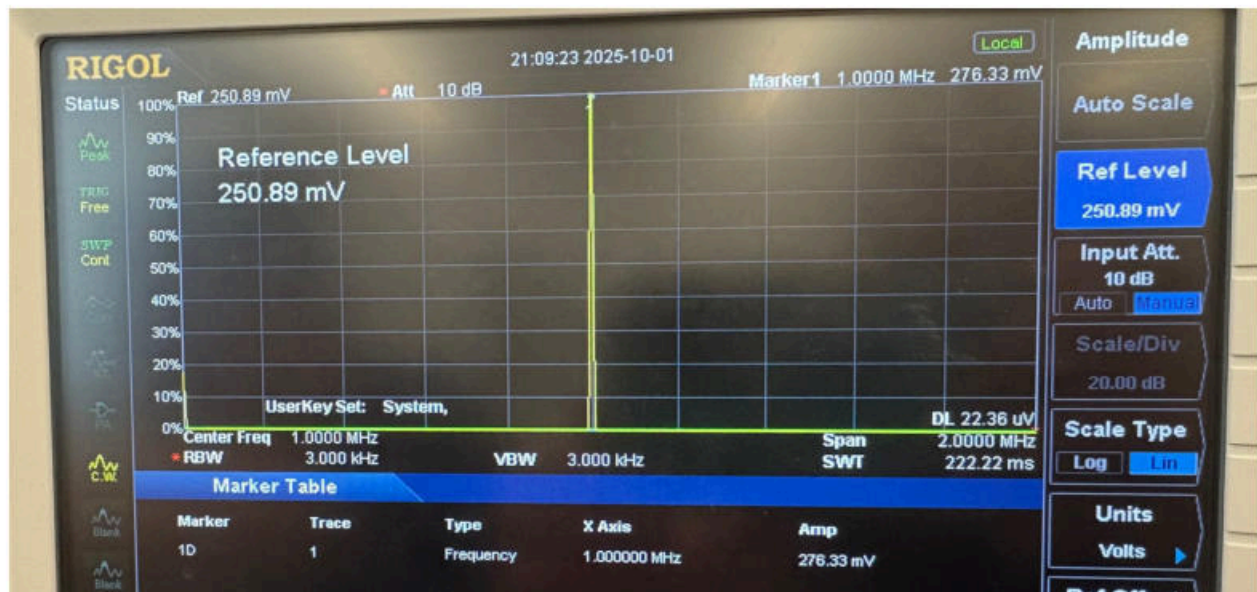


IMG 8.

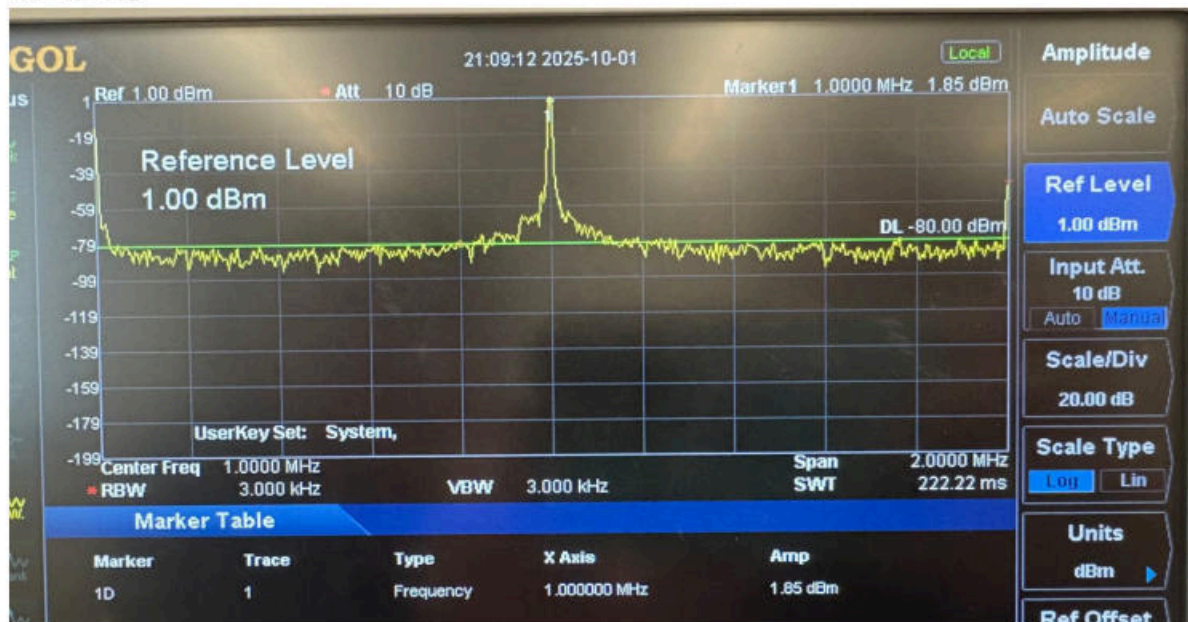


IMG 9

Then, we confirmed that in Linear mode odd harmonics decrease at a rate of $1/n$, so the 5th harmonic will be $1/5$ of the fundamental value. We used the Marker Table (Marker button then down arrow) which displayed multiple signal levels and their frequencies.



IMG 9 Log



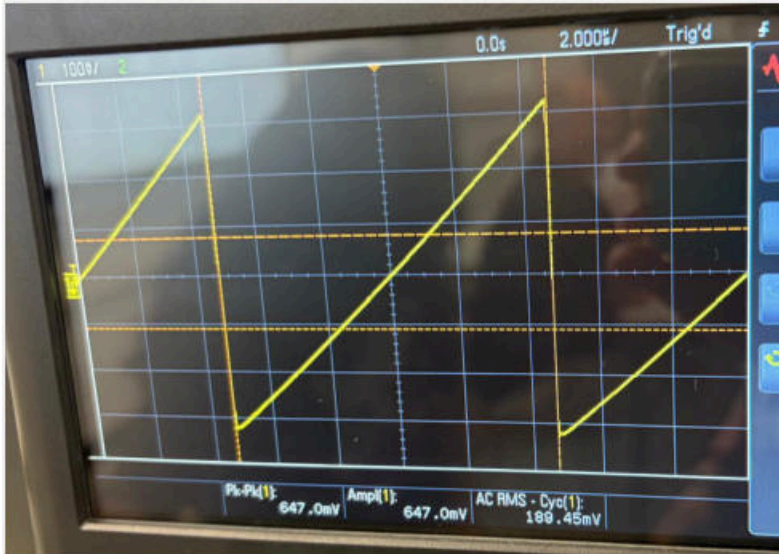
IMG 10 Linear

3. Triangle signal, 1MHz

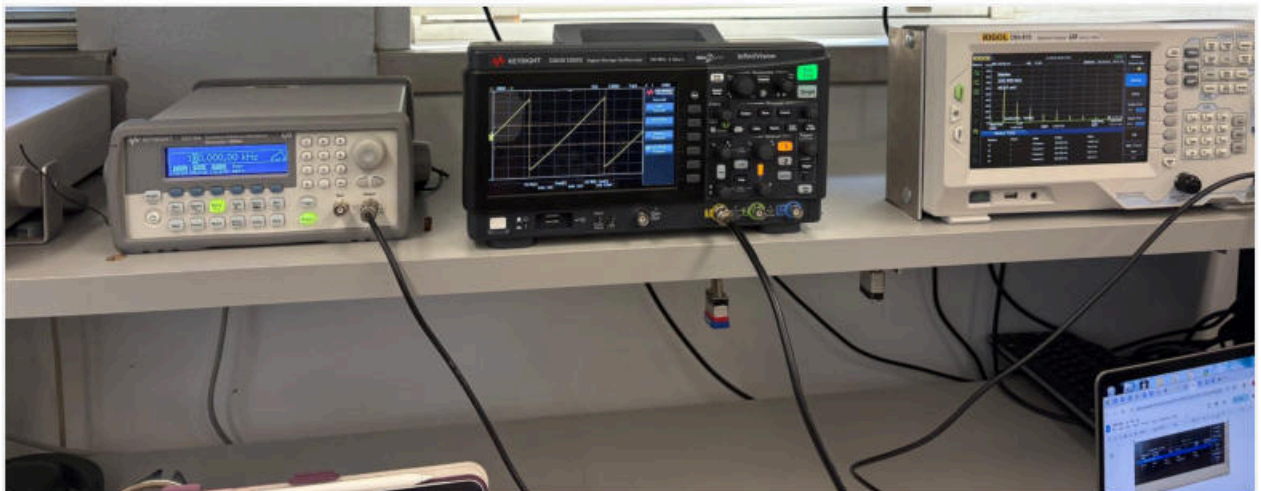
We repeated part 2 with a triangle signal. The maximum possible frequency in this case is 100kHz from the function generator, so we used 100kHz instead of 1MHz.

$$13/5 = 2.6$$

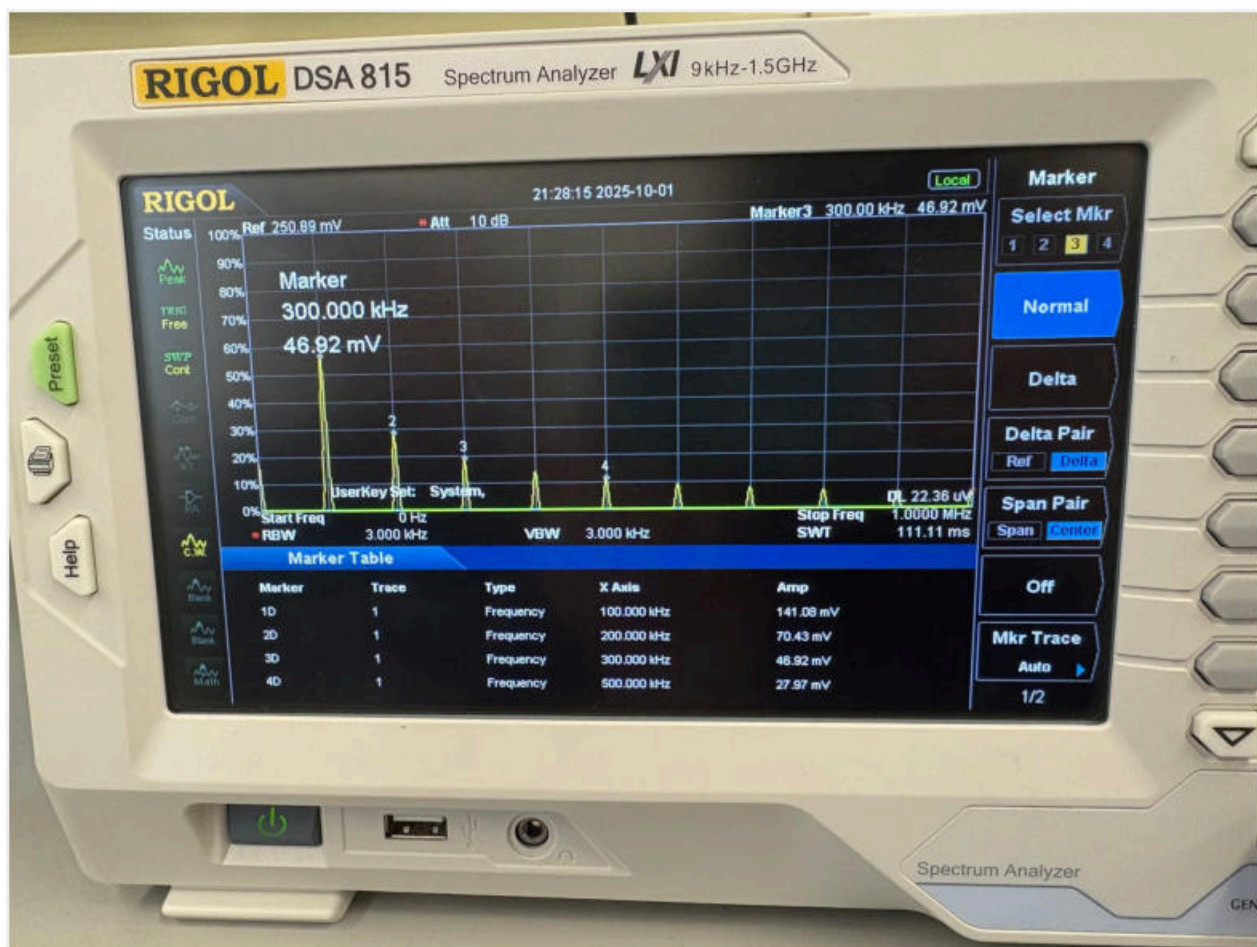
27 mv close



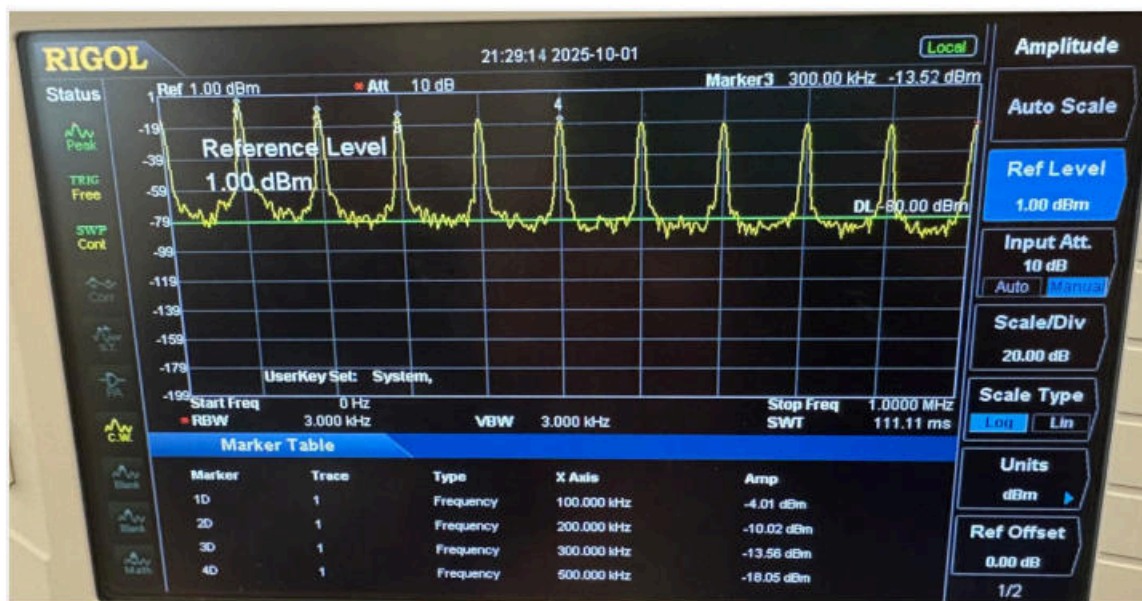
IMG 11



IMG 12.



IMG 13.



IMG 14

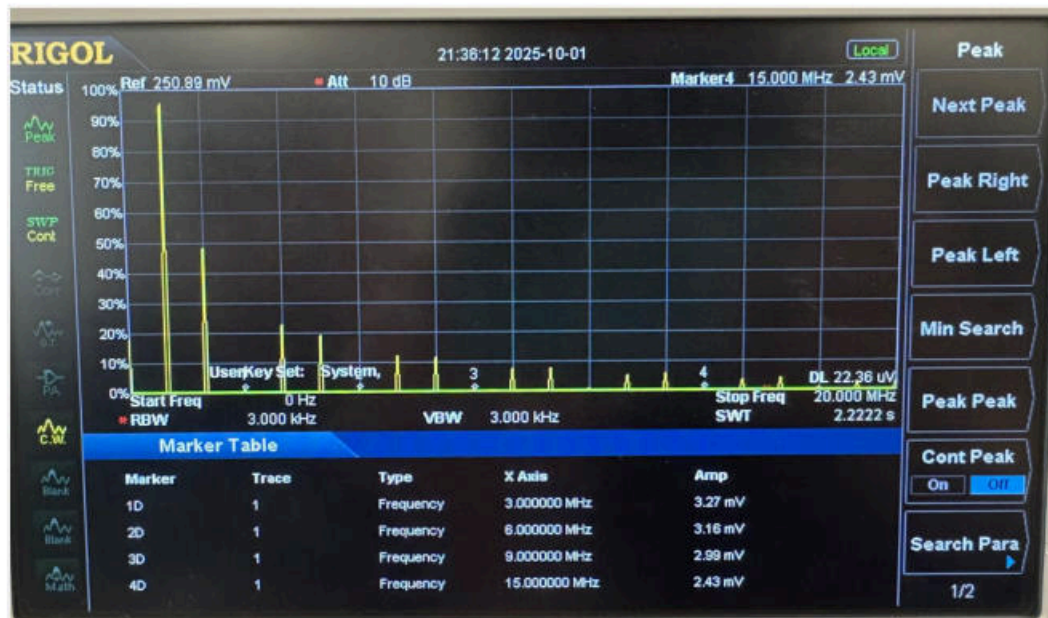
4.. Change to Pulse Signal 1MHz, 33.333% duty cycle, on the Rigol ARB generator adjust the rise and fall times of the signal to a minimum value by changing the Leading and Trailing values to a minimum, 10ns. What do you notice about the 3rd (1/0.333) harmonic level? What about the 6th or 9th?

For the pulse waveform, the edge time was set to the minimum of 20 ns, which is the fastest rise time available on the function generator. This fast rise time creates multiple higher-order harmonics in the frequency spectrum, including the 3rd, 6th, 9th, and 15th harmonics. Switching the Spectrum Analyzer to a 20 MHz span (linear scale) allows us to view the first 20 harmonics clearly on the display. The measured harmonic amplitudes were approximately:

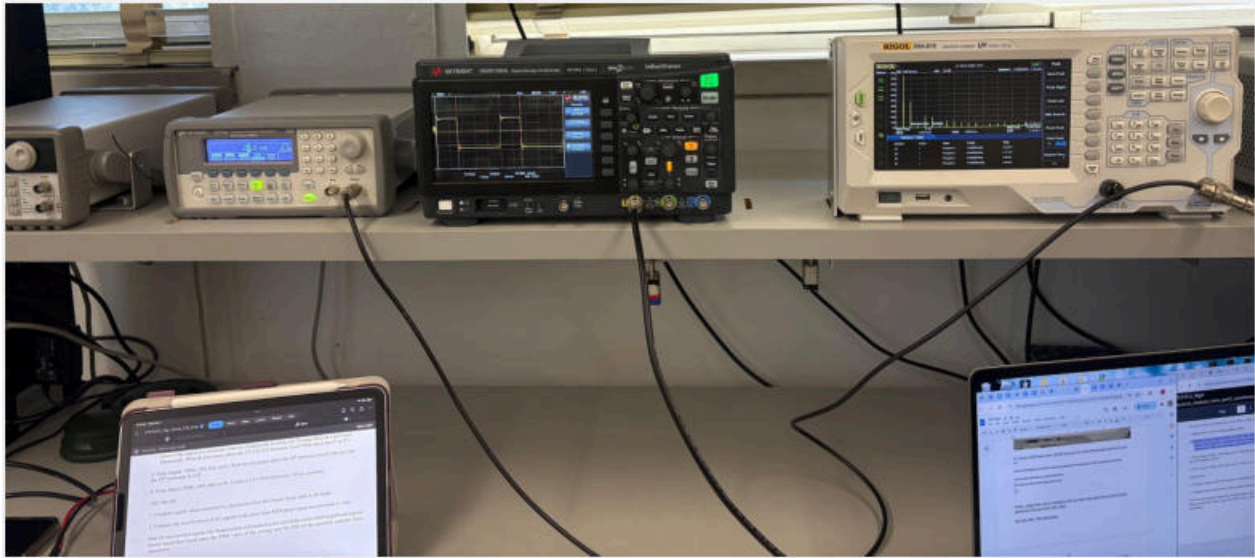
1st harmonic: 239 mV

2nd harmonic: 122 mV

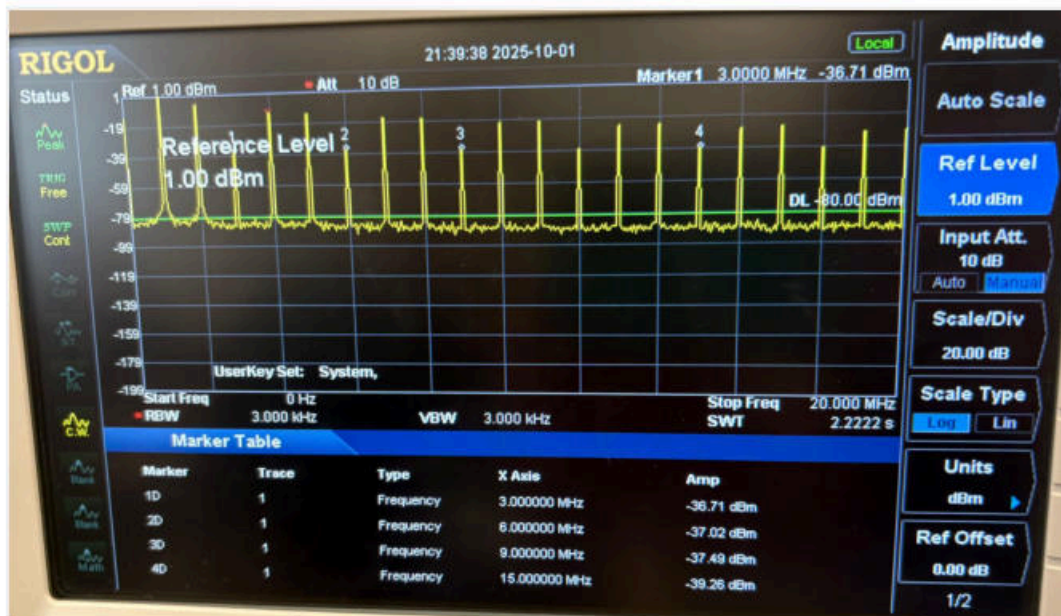
3rd harmonic: 3.27 mV



IMG 15



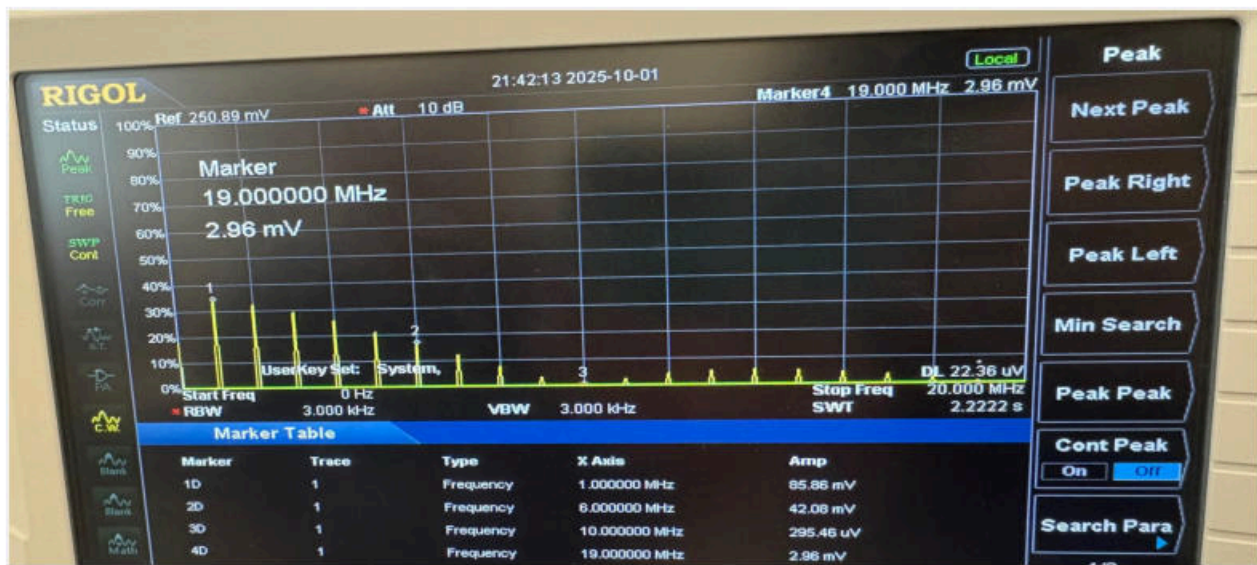
IMG 16



IMG 17 Log

5. Pulse Signal 1MHz, 10% duty cycle

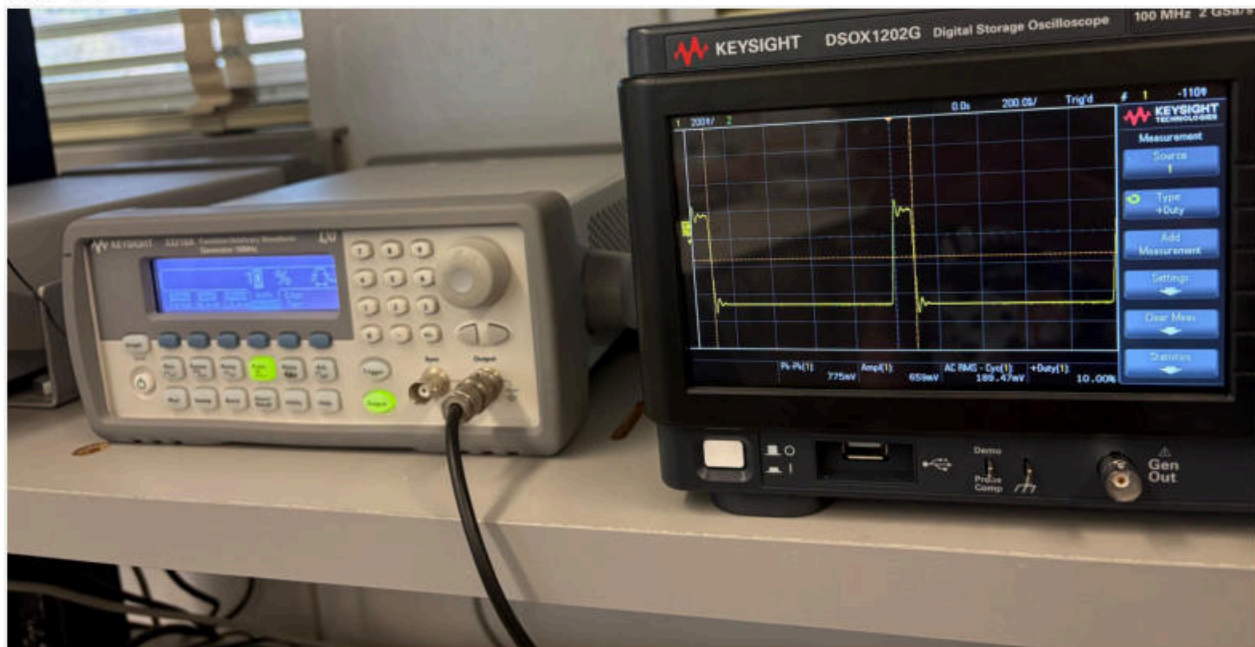
There is no peak at 20 mhz (20th harmonic). There is also no first harmonic (that is the fundamental).



IMG 18

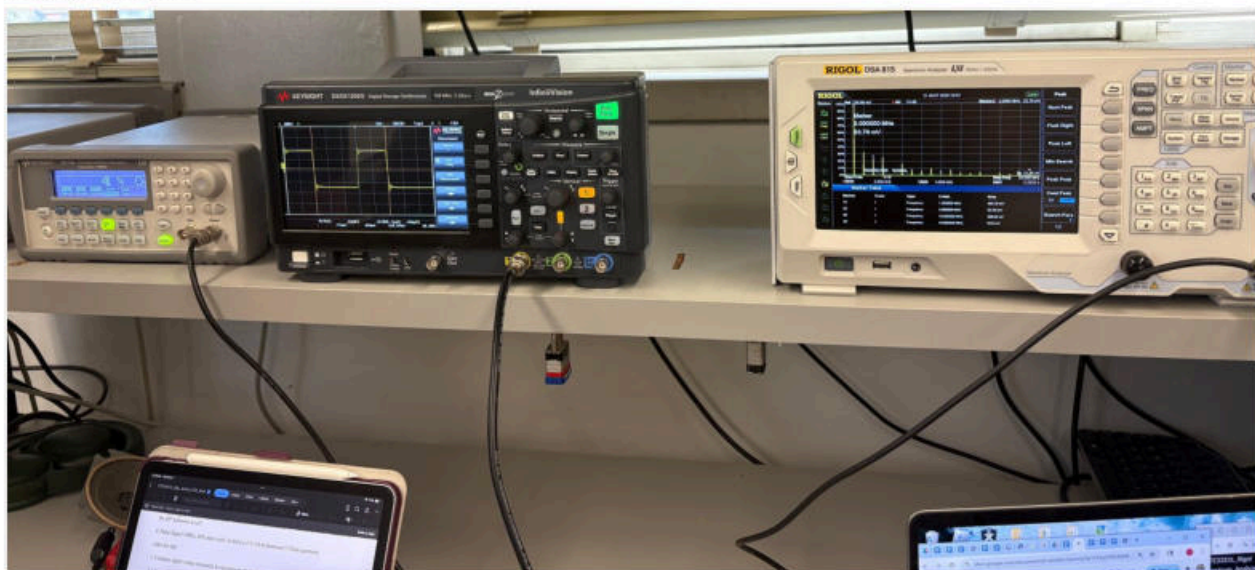


IMG 19

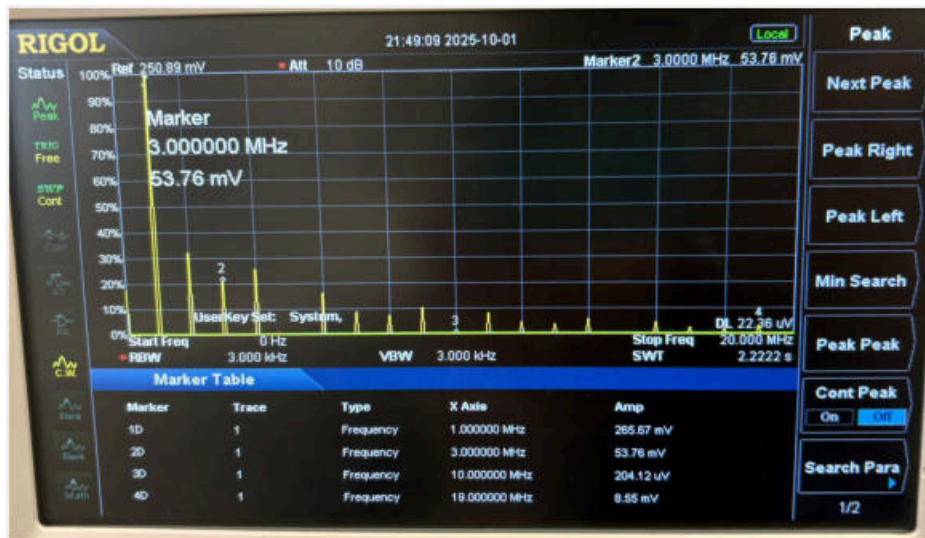


IMG 20.

6. Pulse Signal 1MHz, 40% duty cycle. Is there a 2.5 ($1/0.4$) harmonic? (Trick question)
No there is not



IMG 21.



IMG 22.



IMG 23.

After the lab

1. Compare signal values measured to calculations from the Fourier Series table in the book.
2. Compare the overall power of the signals to the short form RMS power equations presented in class.

Note for uncorrelated signals like fundamentals and harmonics the sum of the power of all (significant) signals should equal that based upon the RMS value of the voltage and the 50 of the spectrum analyzer input impedance.

To verify the accuracy of our measurements we compared the harmonic amplitudes captured by the Spectrum Analyzer to the harmonic predictions from the Fourier Series table for a pulse waveform. Conceptually pulse (and general square-like waves) contains a strong fundamental component and each harmonic becomes smaller as frequency increases. This matched our results: the 1st harmonic was the largest, the 2nd was smaller, and by the 3rd harmonic the amplitude had dropped significantly. Although real-world values were slightly different from the ideal Fourier calculations due to noise, measurement limits, and imperfect rise time the overall trend clearly matched theory — harmonics decrease with harmonic number

We also compared the total power of the waveform to the RMS power calculation learned in class. The RMS equation gives the total signal energy, which includes the fundamental plus all harmonics. Since fundamentals and harmonics are uncorrelated signals, their powers add together independently. In other words:

Total Power = (Power of 1st harmonic) + (Power of 2nd harmonic) + (Power of 3rd ... etc)

We can confirm that when we converted the measured harmonic voltages to power using:

$P = \frac{V_{rms}^2}{50\Omega}$ and by summing these values, expecting that the sum result would be very close to the total power computed directly from the V_{rms} reading of the entire waveform. This confirms the Spectrum Analyzer measurements were valid and followed the expected Fourier Series behavior; a pulse waveform spreads its power across many harmonic frequencies but the fundamental still carries most of the signal strength.

Conclusion

This lab helped us connect theoretical signal analysis with real measurement tools. We confirmed that a pure sine wave produces only one major frequency component while a pulse signal generates many harmonics, matching Fourier Series equation predictions. Using the Spectrum Analyzer we were able to convert voltage measurements into power levels using dB and dBm formulas. We also were reminded how logarithmic scaling makes it easier to observe large power differences. We also verified that total waveform power can be found either from the sum of the harmonic powers or directly from the waveform's V_{rms} value. Through this hands-on exploration we gained important skills in interpreting frequency-domain data, understanding power measurements and using spectrum analysis to evaluate signal behavior.